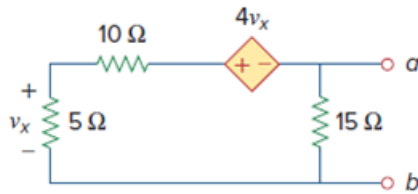


Problem

Obtain the Thevenin equivalent of the circuit in Fig. 4.36.

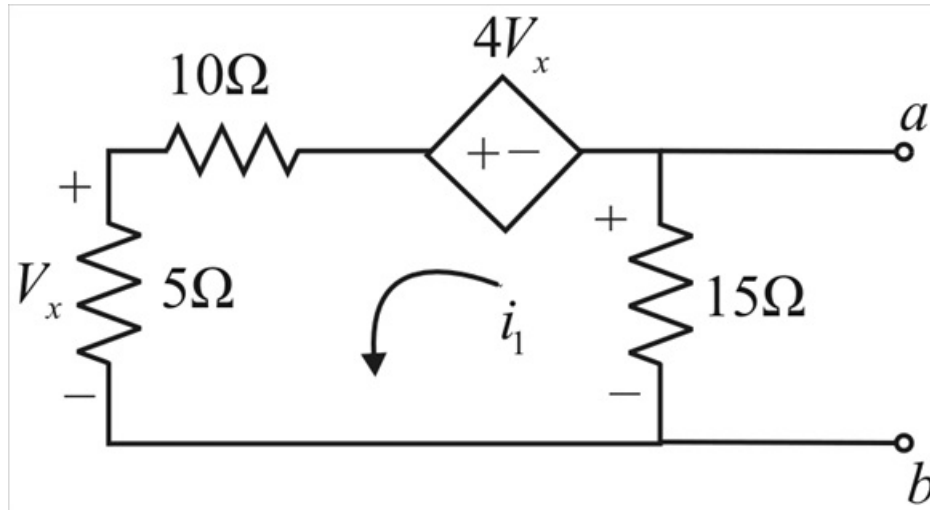
Figure 4.36:



Step-by-step solution

Step 1 of 6

Given circuit is



Step 2 of 6

Determining Thevenin's voltage

By KVL we have

$$-4V_x + 10i_1 + V_x + 15i_1 = 0$$

$$-3V_x + 25i_1 = 0 \dots\dots\dots (1)$$

From the above circuit we can write

$$V_x = 5i_1$$

Now equation (1) becomes

$$-3(5i_1) + 25i_1 = 0$$

$$-15i_1 + 25i_1 = 0$$

$$i_1 = 0$$

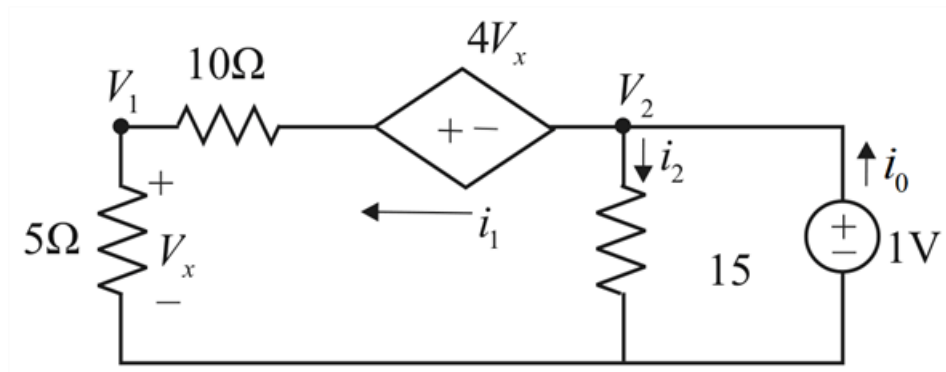
$$V_{Th} = V_{ab} = -15i_1$$

$$V_{Th} = -15 \times 0$$

$$\boxed{V_{Th} = 0V}$$

Step 3 of 6

To determine Thevenin's resistance circuit model is as below

**Step 4** of 6

Applying KCL

$$\frac{V_1}{5} + \frac{V_1 - 4V_x - V_2}{10} = 0$$

From the above circuit

$$V_x = V_1$$

$$\frac{V_1}{5} + \frac{V_1 - 4V_1 - V_2}{10} = 0$$

From the above circuit

$$V_2 = 1$$

$$\frac{V_1}{5} + \frac{-3V_1 - 1}{10} = 0$$

$$\frac{V_1}{5} + \frac{-3V_1}{10} = \frac{1}{10}$$

$$\frac{2V_1 - 3V_1}{10} = \frac{1}{10}$$

$$-V_1 = 1$$

$$V_1 = -1$$

Step 5 of 6

Applying KCL at node V_2

$$i_o = i_1 + i_2$$

$$i_2 = \frac{V_2}{15}$$

$$i_2 = \frac{1}{15}$$

$$i_1 = \frac{V_2 + 4V_x - V_1}{10}$$

$$i_1 = \frac{V_2 + 3V_x}{10}$$

$$i_1 = \frac{1-3}{10}$$

$$i_1 = -0.2$$

$$i_o = -0.2 + \frac{1}{15}$$

$$i_o = -0.1333\text{A}$$

Step 6 of 6

$$R_{th} = \frac{V_2}{i_o}$$

$$R_{th} = \frac{1}{i_o}$$

$$R_{th} = \frac{1}{-0.1333}$$

$$\boxed{R_{th} = -7.5\Omega}$$